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Stat 50

HW #9

3/5/24

Section 2.6

1.

(a) $P(X=1 \text{ and } Y=0)$:-

$P(X=1 \text{ and } Y=0) = 0.17$ From the table

(b) $P(X \geq 1 \text{ and } Y \leq 2)$:-

$P(X \geq 1 \text{ and } Y \leq 2) = 0.17 + 0.23 + 0.06 + 0.14 = 0.60$

(c) $P(X < 1)$:-

$P(X < 1) = P(X = 0) = 0.10 + 0.11 + 0.05 = 0.26$

(d) $P(Y \geq 1)$:-

$P(Y \geq 1) = P(Y=1) + P(Y=2)$

$= 0.11 + 0.23 + 0.14 + 0.05 + 0.08 + 0.06$

$= 0.67$

(e) $P(X \geq 1)$:-

$P(X \geq 1) = P(X=1) + P(X=2)$

$= 0.17 + 0.23 + 0.08 + 0.06 + 0.14 + 0.06$

$= 0.74$

(f) $P(Y=0)$:-

$P(Y=0) = 0.10 + 0.17 + 0.06 = 0.33$

(g) $P(X=0 \text{ and } Y=0)$:-

$P(X=0, Y=0) = 0.10$

(2)

(a) $P(x)$ =

x \ y	0	1	2	$P_x(x)$
0	0.10	0.11	0.05	0.26
1	0.17	0.23	0.08	0.48
2	0.06	0.14	0.06	0.26
(b) $P_y(y)$	0.33	0.48	0.19	1

(c) $\mu_x = 0(0.26) + 1(0.48) + 2(0.26) = 1$

(d) $\mu_y = 0(0.33) + 1(0.48) + 2(0.19) = 0.86$

(e) $\sigma_x = \sqrt{E(x^2) - (E(x))^2}$

$E(x^2) = 0^2(0.26) + 1^2(0.48) + 2^2(0.26)$
$= 1.52$

$\sigma_x = \sqrt{1.52 - 1^2} = 0.721$

(f) $\sigma_y = \sqrt{E(y^2) - (E(y))^2}$

$E(y^2) = 0^2(0.33) + 1^2(0.48) + 2^2(0.19) = 1.24$

$\sigma_y = \sqrt{1.24 - (0.86)^2} = 0.707$

(g) $COV(x,y)$:- $COV(x,y) = E(xy) - E(x)E(y)$

$E(xy) = 0(0)(0.10) + 0(1)(0.11) + 0(2)(0.05)$
$+ 1(0)(0.17) + 1(1)(0.23) + 1(2)(0.08)$
$+ 2(0)(0.06) + 2(1)(0.14) + 2(2)(0.06)$

$E(xy) = 0.91$

$E(x) = (0.26)(0) + 1(0.48) + 2(0.26) = 1$

$E(y) = 0(0.33) + 1(0.48) + 2(0.19) = 0.86$

$Cov(x,y) = 0.91 - (1)(0.86) = 0.05$

(h) ρ_{xy} :-

$$P(x,y) = \frac{Cov(x,y)}{\sigma_x \sigma_y} = \frac{0.05}{(0.721)(0.707)} = \frac{0.05}{0.510047}$$

$P(x,y) = 0.98$

(i) x and y are not independent because cov(x,y) and P(x,y) are not equal to zero.

(4)

(a)

x	y = 0	1	2	3	$P_x(x)$
0	0.15	0.12	0.11	0.10	0.48
1	0.09	0.07	0.05	0.04	0.25
2	0.06	0.05	0.04	0.02	0.17
3	0.04	0.03	0.02	0.01	0.1
(b) $P_y(y)$	0.34	0.27	0.22	0.17	1

(c) $P(x=0, y=0) = 0.15 \neq P_x(0) P_y(0) = (0.48)(0.34) = 0.1632$

$P(x=0, y=0) \neq P_x(0) P_y(0)$ ∴ They are not independent.

(d) μ_x :-

$$\mu_x = 0(0.48) + 1(0.25) + 2(0.17) + 3(0.1)$$
$$= 0.89$$

μ_y :-

$$\mu_y = 0(0.34) + 1(0.27) + 2(0.22) + 3(0.17)$$
$$= 1.22$$

(e) σ_x :-

$$E(x^2) = 0^2(0.48) + 1^2(0.25) + 2^2(0.17) + 3^2(0.1)$$
$$= 1.83$$

$$\sigma_x = \sqrt{1.83 - (0.89)^2} = 1.01877$$

σ_y :-

$$E(y^2) = 0^2(0.34) + 1^2(0.27) + 2^2(0.22) + 3^2(0.17)$$
$$= 2.68$$

$$\sigma_y = \sqrt{2.68 - (1.22)^2} = 1.09160$$

(f) $Cov(x,y)$:- $E(xy) - E(x)E(y)$

$$E(x,y) = 1(0.07) + 2(0.05) + 3(0.04) + 2(0.05) + 4(0.04) + 6(0.02) + 3(0.03) + 6(0.02) + 9(0.01)$$
$$= 0.97$$

(g) $Cov(x,y) = 0.97 - (0.89)(1.22) = -0.1158$

(g) $P(x,y) = \dfrac{Cov(x,y)}{\sigma_x \sigma_y} = \dfrac{-0.1158}{(1.01877)(1.09160)} = -0.10413$

(a)

X \ Y	0	1	2	3	4	$P_X(x)$
0	0.03	0.01	0.02	0.01	0.01	0.08
1	0.09	0.05	0.03	0.03	0.01	0.26
2	0.09	0.05	0.07	0.04	0.02	0.27
3	0.06	0.04	0.07	0.04	0.02	0.23
4	0.04	0.03	0.04	0.03	0.02	0.16
(b) $P_Y(y)$	0.31	0.18	0.28	0.15	0.08	1

(c) No, $P_{XY}(0,0) = 0.03$ but $P_X(0)P_Y(0) = 0.0248$

$P(X=0, Y=0) \neq P_X(0)P_Y(0)$

(d) M_X :-

$M_X = 0(0.08) + 1(0.26) + 2(0.27) + 3(0.23) + 4(0.16)$

$= 2.13$

M_Y :-

$M_Y = 0(0.31) + 1(0.18) + 2(0.28) + 3(0.15) + 4(0.08)$

$= 1.57$

(e) σ_X :-

$\sigma_X = \sqrt{E(X^2) - (E(X))^2}$

$E(X^2) = 0^2(0.08) + 1^2(0.26) + 2^2(0.27) + 3^2(0.23) + 4^2(0.16)$

$E(X^2) = 5.97$

$\sigma_X = 1.1971$

σ_Y :- $\sigma_Y = \sqrt{E(Y^2) - (E(Y))^2}$

$E(Y^2) = 0^2(0.31) + 1^2(0.18) + 2^2(0.28) + 3^2(0.15) + 4^2(0.08)$

$= 3.93$

$\sigma_Y = 1.2845$

(8) $cov(x,y)$:-

$$E(xy) = 1(0.05) + 2(0.08) + 3(0.03) + 4(0.01) + 2(0.05) + 4(0.07) + 6(0.04) + 8(0.02) + 3(0.04) + 6(0.07) + 9(0.04) + 12(0.02) + 4(0.03) + 8(0.08) + 12(0.03) + 16(0.02)$$

$$cov(x,y) = E(xy) - E(x)E(y)$$
$$= E(x,y) - (2.13)(1.51)$$
$$= 0.1687$$

(9) $P(x,y)$:-

$$P(x,y) = \frac{cov(x,y)}{\sigma_x \sigma_y} = \frac{0.1687}{(1.1971)(1.2845)}$$

$$P(x,y) = 0.1065$$

(10)

(a) $E(xy) = 41/6$

$E(x)E(y) = \int_4^5 \left(\int_1^6 \frac{1}{6}(x+y)\,dx \right) y\,dy$

$\int_1^6 \left(\int_4^5 \frac{1}{6}(x+y)\,dy \right) x\,dx$

$$cov(x,y) = E(x,y) - E(x)E(y)$$
$$= -0.000193$$

(b) P_{xy} :-

$$\frac{cov(x,y)}{\sigma_x \sigma_y} = \frac{-0.000193}{0.083189155}$$

$$P(x,y) = -0.00232$$

(18)

(b) $f_x(x)$:-

$$f_x(x) = \int_0^1 \frac{3}{2}(x^2+y^2)\,dx = \frac{3}{2}\left(\frac{x^3}{3} + xy^2\right)\Big|_0^1$$

$$= \frac{3}{2}\left(\frac{1}{3} + y^2\right) - 0 = \frac{3}{2}\left(\frac{1+3y^2}{3}\right)$$

$$= \frac{1}{2}(1+3y^2)$$

$f_y(y)$:-

$$f_y(y) = \int_0^1 \frac{3}{2}(x^2+y^2)\,dy = \frac{3}{2}\left(x^2y + \frac{y^3}{3}\right)\Big|_0^1$$

$$= \frac{3}{2}\left(x^2 + \frac{1}{3}\right) - 0 = (3x^2+1)\left(\frac{1}{2}\right)$$

$$f_y(y) = \frac{1}{2}(3x^2+1)$$

(c) $\int_0^1 \frac{1}{2}(1+3y^2)\,dy = \frac{1}{2}\left(y + \frac{3y^3}{3}\right)\Big|_0^1 = 1$

$\int_0^1 \frac{1}{2}(3x^2+1)\,dx = \frac{1}{2}(x^3+x)\Big|_0^1 = \frac{2}{2} = 1$

Yes, x and y are independent because $f_x(x)$ and $f_y(y) = 1$.

(19)

(a) $COV(X,Y)$:-

$E(XY) = 3/8$

$E(X)\ E(Y) =$

$COV(X,Y) = E(XY) - E(X)\,E(Y)$

$= -0.015625$

(b) $P(X,Y)$:-

$P(X,Y) = P_X(x) \cdot P_Y(y)$

$P(X,Y) = -0.2055$

(c) $E(Y|X=0.5)$

$$f_{Y|X}(y|0.5) = \begin{cases} \frac{12y^2+3}{7} & 0 < y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(21)

(a) $$f(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

$$f(x,y) = \begin{cases} e^{-x-y} & x>0 \text{ and } y>0 \\ 0 & \text{otherwise} \end{cases}$$

(c) M_x :-